Fiscal Year:	FY 2022	**Task Last Updated:**	FY 06/20/2022
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Project Title:	Complex Plasma under Microgravity, Utilizing the International Space Station Experiment “PK-4” and Beyond		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:	National Science Foundation	**TechPort:**	No
Human Research Program Elements:	None		
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Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2016 Physical Sciences NNH16ZTT002N: Fundamental Physics – NASA/NSF Partnership on Science of Dusty Plasmas: Utilizing the PK-4 Facility on board the ISS
Start Date:	04/15/2017	**End Date:**	12/31/2023
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	4	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	6
No. of Bachelor's Candidates:	6	**Monitoring Center:**	NASA JPL
Contact Monitor:	Callas, John	**Contact Phone:**	
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Flight Program:	ISS		
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Thomas, Edward Ph.D. (Auburn University) Williams, Jeremiah Ph.D. (Wittenberg University)		
Grant/Contract No.:	JPL 1571699		
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Performance Goal Text:			

Task Description:

Complex plasmas are four-component systems that contain electrons, ions, neutral atoms, and charged microparticles (i.e., the dust particles). Depending upon the experimental conditions, the microparticles in these systems can show solid-like, liquid-like, or gas-like properties. However, ground-based experiments are subject to gravitational forces that often dominate the behavior of the systems under investigation leading to sedimentation and anisotropic effects. Therefore, in order to study large-scale complex plasmas and the influences of the weaker forces that are masked by the gravitational force, experiments have to be performed under “microgravity” conditions. Over the past decade, several complex plasma experiments have been performed using the International Space Station (ISS). The latest of these research instruments, “Plasma Kristall 4” (PK-4), is the first microgravity laboratory that is optimized to study fluid phenomena and fundamental effects such as particle charging, screening, and the influence of a directed ion flow. Additionally, since PK-4 is a multi-user facility, we have a unique opportunity to use this instrument to study ion/particle and particle/particle interactions that we otherwise would not be able to address in ground-based experiments.
The intellectual merit of this research project is centered on three major research topics.

Topic 1: Charging and interaction of microparticles in the plasma. The most important measurement in a complex plasma is the measurement of the charge on the particles. However, direct experimental measurement of the charge is difficult, because it usually relies on additional assumptions about experimental parameters such as the plasma electric field and on the validity of an assumed particle interaction type. The microgravity environment of PK-4, where the dust is suspended in the bulk plasma, allows experiments that can potentially lead to a higher accuracy in the particle charge determination.

Topic 2: Ion-dust interactions and the thermal properties of dusty plasmas. A key observation in complex plasmas is the unusually high dust kinetic temperature. It has been postulated that ion-dust interactions are largely responsible for heating the dust particles. The microgravity environment of the PK-4 laboratory, in combination with the tools to actively manipulate the ions in the plasma, provides a unique platform for studying the transfer of energy between ions and the microparticles in the plasma.

Topic 3: Particle transport and heating in waves. The low charge to mass ratio of the dust leads to characteristic frequencies for the dust component that are on the order of a few Hz. This, coupled with the finite size of the dust grains, allows researchers a unique opportunity to directly image plasma physics phenomena and particle transport at the kinetic level. In the microgravity environment of PK-4, it will be possible to measure the transport properties of particles during their initial injection into the bulk plasma, follow the particles as they form collective structures (e.g., waves), and study how ion flow modifies the thermal properties of these waves.

This proposed research project has several opportunities to have a broader impact: (1) through on- and off-campus outreach activities and the training of undergraduate and graduate students, this project will cultivate the next generation of plasma scientists; (2) through the development of new ground-based laboratory instruments (e.g., plasma sources, diagnostics) that will complement the proposed research and contribute to the development of next-generation dusty plasma microgravity experiments; and (3) through the development of new, advanced software tools for image analysis and particle tracking that will support the proposed studies and be offered as community-wide research tools.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

The research performed under this project falls in the main category of fundamental research. Often, findings obtained by performing fundamental research show their true impact for the society only after many years. However, the study of dusty/complex plasma can benefit the overall research infrastructure in many ways already in the timely vicinity of any discovery made, mainly as a result of the strong interdisciplinary nature of this research. Studying complex plasma topics such as wave phenomena, heating, the spatial separation of phases, their interaction or the transition between phases such as from fluids to crystals all being analyzed down to the scale of individual particles can lead to fundamental finding that might be immediately applied to many other research fields such as solid state physics, fluid physics, plasma physics, astrophysics. As the specific research performed within the scope of this project make use of the microgravity (free-fall) environment supported by the national laboratory in space, the International Space Station (ISS), further utilizing an experiment that is built by ESA, highly international cooperation activities are a direct result. It is likely that the effort to work on the same fundamental questions in a cooperative acting international team will ultimately benefit each involved nation also beyond just the research. Some aspects of the specific research topics within the scope of this project involves the development of universal analysis software, hardware or analysis tools that can be used under different research situations too and some of the more specific technical tools that are developed such as a plasma impedance probe might even help to overall improve technology that can be used for industrial purposes. Understanding the behavior of dust in a plasma environment, especially in a low gravity environment, is further highly advantages for example in mitigating the effect and connected hazards of dust for future, manned space mission to the Moon and/or Mars. Our activities on the described project indeed triggered follow up activities towards lunar dust mitigation research and related technology developments. As part of the general research mission, we also train the next generation of engineers and scientist who can in the course of their future career produce a large benefit for the overall economic situation of the nation and beyond.

The report given here is comprehensive with respect to the initial projects research agreement with the Jet Propulsion Laboratory (JPL) that started in 2017 and has been conducted till recently, in December 2022, including a granted no cost extension from April 2022 to December 2022. The report on major follow-up activities under this project that have started in Mid 2022 under a supplemental agreement with JPL and that has expanded the scientific scope (research topics) we have been investigating since mid 2022 is included.
The project “Complex Plasma under Microgravity, Utilizing the International Space Station Experiment PK-4 and Beyond” started 04/16/2017 with the work agreement “RSA-JPL-1571699”. The latter agreement was extended once from 04/16/2021 till 12/15/2021. The research under the agreement was conducted in cooperation between Auburn University and Wittenberg University. In addition, NSF supported the work of undergraduate students at Auburn for this project under the contract NSF-PHY-1740784. The overall mission under the research agreement has been to perform “Basic Research” in the field of dusty/complex plasma by utilizing the PK-4 experiment to advance the knowledge in the field of complex plasma and to advance potential activities beyond PK-4 for “microgravity” based research in the field. The activities were organized in the proposed topics:

Topic T1: Charging and interaction of micro-particles in a plasma,

Topic T2: Ion-dust interaction and the thermal properties of dusty plasmas,

Topic T3: Particle transport and heating in waves,

Topic TX: Future space experiments, administrative tasks and analysis tool developments.

With the help of two PhD students, we have studied different techniques to study the particle charging of dust in a plasma environment (Topic T1). One technique involved the evaluation of the overall charge balance in a complex plasma. For the latter we first developed a plasma impedance / Langmuir probe combination systems to determine the electron and ion densities in the plasma. The reminding balance of the charge is supposed to be mainly supported by the dust in the system and as such the individual dust charge can be estimated by the combine measurement as well as a dust density estimation. A different approach has been taken by a second PhD student who studied the effect of a volumetric force field such as gravity or Lorenz force on a flowing or sedimenting complex plasma. The performed simulations and the constructed theoretical framework predicts that the particle charge can be derived from observable, predicted occurring dust density gradients. Thus a detailed observation of the dust particle positions in an experiment can be used to derive the dust charge under some suitable assumptions on the nature of the particle interaction.

Topics T2/T3 have been studied by a third graduate students as well as multiple undergraduate students both at Auburn University and Wittenberg University. These studies made highly use of the PK-4 space experiment as well as simulations and theory. The Auburn graduate student several times even was leading the organization and conduction of the proposed experiment in the international cooperative environment that involved space agencies / institutions such as NASA, ESA, DLR, CADMOS as well our national partner universities for the individual experiments. The students and supervising PIs highly made use of the complex plasma features that allows to study the dynamics and structure of the system down to the position and motion of the individual particle, the “quasi atoms” of the system. Studying the system on the quasi-atomic level allowed to gain new knowledge of complex plasma dynamics and the role of the ion flow in the underlying dc discharge on the interaction between the particles. All three PhD works are in the process of conclusion.

The topic TX has been mainly studied by the PI as well as one of the Auburn graduate students utilizing experimental setups and contributing to their development for experiments that may become follow-up experiment facilities for PK-4. As a result of our effort, we made significant process in establishing an international cooperative environment for complex plasma research under microgravity conditions with significant contribution from the US. We succeeded in acquiring a leading role in the development and definition of potential next generation space experiment for complex plasma such as COMPACT, and we already triggered development activities for experiments beyond the latter. Further, we started activities as part of different projects to apply our gained knowledge for dust mitigation studies that might become highly relevant in future manned explorations of the Moon and/or Mars.

Task Progress:

Over the course of the project, a very fruitful cooperation environment has been build up to perform experiments in a team of multiple institutions. National partners such as Baylor University and University of California (UCSD) have been included, as well as several international partners, such as University of Giessen (Germany), the German Aerospace Center (DLR e.V.), University of Greifswald (Germany) and initially also to the Institute for High Energy Densities (IHED) from the Russian Academy of Science (RAS) in Moscow. Under the national/international cooperation multiple experiment campaigns were performed utilizing the International Space Station (ISS) experiment PK-4 (“Plasmakristall-4”). In detail, we participated within this scope of the initial research agreement in PK-4 campaigns (Campaign number, date, title of experiment):

• C7 (June 2019) – “Measurements of charging, energy, and dust chains in PK-4”,

• C9 (Feb 2020) – “Energy Transport in Multi-Chain Dusty Plasma”,

• C11 (Mar 2021) – “Stopping Power”,

• C12 (Jun 2021) – “Follow-up Experiment on Energy Dissipation”,

• *C14 (Feb 2022) – “Dust Cloud Structuring and Dynamics Induced by Active Compression of the System”,

• *C15 (under prep.) – Follow-up experiment to *C14.

The additional experiments (marked with *) have been performed and are in preparation, supported by a supplemental research agreement with JPL (RSA 1667433) that has expended the scientific scope of our activities under this project towards the study of the structure and dynamic of large scale complex plasma system. It has to be noted that experiments with PK-4 were not possible for us in the initial years of the project as the PK-4 operation schedule was delayed due to technical issues with the setup and because of a variety of initial experiments that had to be performed by the PK-4 core group (for example a set of commissioning experiments). To mitigate the delay in our PK-4 access, experiments using ground base PK-4 breadboard setups have been utilized. We also performed a variety of experiments utilizing the breadboard setup of the potential next generation space experiment “COMPACT” (especially related to Topic TX). The latter setup was located at the German Aerospace Center in Germany. As part of the recently added research topic to study large scale complex plasma systems we developed an experiment scenario to study the structural transition and dynamics during the process of a complex plasma cloud while it is actively compressed. As this experiments made use of yet untested experiment capabilities of PK-4 we performed two experiment sessions on the ground reference module to test new manipulation techniques in the fall of 2022. In February 2022 we performed a final training experiment with the ground reference module before executing the experiment with the PK-4 setup aboard the ISS a few weeks later. We are still waiting to receive the high resolution experiment data. The initial quick-look data, a few seconds of experimental data, indicate a successful experiment. The observation already triggered the planning of a follow-up experiment intended to be executed on one of the next available campaigns.

Similar to probably many other groups, also we were impacted by the COVID pandemic. Due to travel restrictions we missed multiple opportunities for cooperative research with our European colleges. Namely, a planned sabbatical of the PI under this project to work at DLR had to be canceled and as a result also the participation in two parabolic flight campaigns, one featuring the PK-4 ground experiment and one the COMPACT breadboard setup. The data delivery from the experiment has been significantly delayed too, so that much of our analysis had to be performed with a small set of “quick-look” data or data from the first two campaigns. We adapted to the situation by shifting priorities and utilizing local experimental setups for part of the research when PK-4 and the breadboard setups were not accessible or we concentrated on molecular dynamic (MD) simulations and enhancing our theoretical modeling to tackle the research

	topics. On the other hand, the enhanced video communication technology capabilities pushed by the need of the COVID situation, helped us to enhance overall scientific collaborations. The latter played a significant role when we hosted an international workshop in 2021, “NASA BPS Fundamental Physics Decadal Survey Workshop (Virtual) – From Dust in Space and on Solar System Bodies to Dusty/Complex Plasma studies under reduced gravitational Influence” together with the University of Maryland. Also as a result of travel restrictions, the participation in PK-4 experiments had to be made more flexible. Instead of traveling to CADMOS (France) to participate in the mission, there has been the need to allow PIs to work remotely on the PK-4 experiments. Together with NASA/CADMOS/CNRS and ESA we tested and established a way to allow the participation in PK-4 missions remotely from the office of the PI or Co-I. The now several times tested experimental access methods can be a base model for future participation in PK-4 and for international space experiment cooperation beyond PK-4 such for the potential upcoming COMPACT project or potential microgravity research activities of other groups that requires the involvement of multiple, international space agencies.
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